STPM Physics · Semester 3

DEFINITION LIST

Oscillations, Waves & Modern Physics · Chapters 20–26. 194 definitions, laws and statements; key words in bold. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 20 — OSCILLATIONS

20.1 Characteristics of Simple Harmonic Motion

1. **Simple harmonic motion** is the motion of a particle whose acceleration is directly proportional to its displacement from a fixed (equilibrium) point and is always directed towards that point.
2. **Amplitude** is the maximum displacement of the particle from the equilibrium position (m).
3. **Equilibrium position** is the point at which no resultant force acts on the body performing SHM; displacement is measured from this point.
4. **Period** is the time taken for one complete oscillation; **frequency** is the number of complete oscillations per unit time.
5. **Angular frequency** is the rate of change of phase angle, $\omega = 2\pi f = \frac{2\pi}{T}$ (rad s^{-1}).
6. **Phase difference** is the fraction of a cycle by which one oscillation is ahead of or behind another, expressed in radian or degree.
7. In SHM the restoring force is directly proportional to the displacement and opposite in direction to it. Velocity leads displacement by $\frac{\pi}{2}$ rad; acceleration is in antiphase (π rad) with displacement.

20.2 Kinematics of Simple Harmonic Motion

8. **Displacement** in SHM is a sinusoidal function of time; for a particle starting from the equilibrium position, $x = A \sin \omega t$.
9. Velocity is the rate of change of displacement, $v = \frac{dx}{dt}$, and acceleration is the rate of change of velocity, $a = \frac{dv}{dt} = -\omega^2 x$.
10. The speed at displacement x is given by $v = \pm\omega\sqrt{A^2 - x^2}$; the speed is zero at $x = \pm A$ and maximum at $x = 0$.

11 The period of SHM is independent of the amplitude.

20.3 Energy in Simple Harmonic Motion

12 **Total energy** of a particle in undamped SHM is constant and is equal to its maximum kinetic energy (at the equilibrium position) or its maximum potential energy (at the extreme positions).

13 Kinetic energy is maximum at $x = 0$ and zero at $x = \pm A$; potential energy is zero at $x = 0$ and maximum at $x = \pm A$.

14 Energy is continuously interchanged between kinetic and potential forms with the total remaining constant when there is no damping.

15 The total energy is directly proportional to the square of the amplitude.

20.4 Systems in Simple Harmonic Motion

16 A **mass–spring system** performs SHM because the restoring force on the mass is directly proportional to its extension and is directed towards the equilibrium position (Hooke's law, $F = -kx$).

17 A **simple pendulum** performs SHM only for small angular displacement ($\sin\theta \approx \theta$ in radian, up to about 10°); the restoring force along the arc is $mg\sin\theta \approx mg\theta$.

18 The period of a simple pendulum is independent of the mass of the bob and of the amplitude (for small angles).

19 The period of a mass–spring system is independent of the amplitude and, for a vertical spring, of g.

20.5 Damped Oscillation

20 **Damped oscillation** is an oscillation in which the amplitude decreases with time because energy is dissipated (as heat) by resistive forces such as air resistance or friction.

21 **Light (under) damping**: the system oscillates with amplitude decreasing exponentially with time; the period is slightly longer than that of the free oscillation (almost unchanged when the damping is light).

22 **Critical damping**: the system returns to the equilibrium position in the shortest possible time without oscillating.

23 **Heavy (over) damping**: the system returns to the equilibrium position slowly without oscillating.

24 Explain damping: the resistive force does negative work on the oscillator, so the total mechanical energy decreases and the amplitude decreases.

20.6 Forced Oscillation and Resonance

25 **Free oscillation** is the oscillation of a system without any external force acting on it after the initial disturbance; no energy is lost, so the amplitude is constant and the system oscillates at its natural frequency.

26 **Forced oscillation** is an oscillation of a system driven by an external periodic force; it oscillates at the frequency of the driving force.

27 **Resonance** occurs when the driving frequency is equal to the natural frequency of the system, giving maximum amplitude of oscillation (maximum energy transfer to the system).

28 Effect of damping: increasing damping reduces the amplitude at resonance and makes the resonance curve broader (less sharp); the peak shifts slightly to lower frequency.

29 At resonance the driving force leads the displacement by $\frac{\pi}{2}$ rad.

CHAPTER 21 — WAVE MOTION

21.1 Progressive Waves

1 A **progressive (travelling) wave** is a wave whose profile moves along the medium; it transfers energy from one point to another through a medium (or vacuum) by oscillations of the particles (or fields), without transferring matter.

2 **Wavelength** is the distance between two successive points that are in phase on the wave (the distance travelled by the wave in one period).

3 A **wavefront** is the locus of points which vibrate in phase.

4 **Amplitude** is the maximum displacement of a particle from its equilibrium position; **period** is the time for one complete oscillation of a particle.

5 **Wave speed** is the distance travelled by the wavefront per unit time.

6 **Phase difference** between two points on a progressive wave depends on their separation along the direction of travel.

7 In a transverse wave the oscillations are perpendicular to the direction of propagation; in a longitudinal wave they are parallel to it.

21.2 Wave Intensity

8 **Intensity** of a wave is the rate of energy transferred per unit area perpendicular to the direction of propagation (power per unit area), unit W m^{-2}.

9 Intensity is directly proportional to the square of the amplitude (for the same frequency and medium).

10 For a point source radiating uniformly in all directions with no absorption, intensity is inversely proportional to the square of the distance from the source (inverse square law).

11 Explain the inverse square law: the same power spreads over the surface of a sphere of area $4\pi r^2$, which increases as r^2.

21.3 Principle of Superposition

12 **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.

13 **Coherent sources** are sources that emit waves of the same frequency with a constant phase difference.

14 **Constructive interference** occurs when the waves meet in phase: the path difference is $n\lambda$ ($n = 0, 1, 2, \ldots$), giving the maximum resultant amplitude.

15 **Destructive interference** occurs when the waves meet in antiphase: the path difference is $(n + \frac{1}{2})\lambda$, giving the minimum resultant amplitude.

16 A stable, observable interference pattern requires coherent waves of similar amplitude and the same frequency.

21.4 Standing Waves

17 A **standing (stationary) wave** is a wave whose profile does not move; it is formed by the superposition of two progressive waves of the same frequency and amplitude travelling in opposite directions.

18 A **node** is a point of zero amplitude (permanent destructive interference); an **antinode** is a point of maximum amplitude.

19 Adjacent nodes (or adjacent antinodes) are separated by $\frac{\lambda}{2}$; a node and the adjacent antinode are separated by $\frac{\lambda}{4}$.

20 Differences from a progressive wave: a standing wave does not transfer energy along the medium; its amplitude varies with position (zero at nodes); particles between two adjacent nodes oscillate in phase, and particles in adjacent segments oscillate in antiphase.

21 For a string fixed at both ends there is a node at each end; only wavelengths with an integral number of half-wavelengths fit on the string.

21.5 Electromagnetic Waves

22 **Electromagnetic waves** are transverse waves consisting of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation; they do not need a material medium.

23 All electromagnetic waves travel at the same speed, $c = 3.00\times10^{8}$ m s^{-1}, in vacuum; they differ in frequency and wavelength.

24 In an electromagnetic wave, the electric and magnetic fields oscillate in phase.

25 Order of the spectrum, from longest wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays.

26 Electromagnetic waves can be polarised, reflected, refracted, diffracted and show interference (properties of transverse waves).

▶ CHAPTER 22 — SOUND WAVES

22.1 Propagation of Sound Waves

1 **Sound waves** are longitudinal mechanical waves in which the particles of the medium oscillate parallel to the direction of propagation, forming alternate regions of compression and rarefaction.

2 A sound wave needs a material medium; it cannot travel through a vacuum, and there is no net flow of the medium from source to listener.

3 The **speed of sound** depends on the elasticity and density of the medium: $v = \sqrt{\frac{B}{\rho}}$ in a fluid and $v = \sqrt{\frac{E}{\rho}}$ in a solid, so v is greater in solids than in liquids and gases.

4 In an ideal gas $v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{\gamma p}{\rho}}$, so v is proportional to $\sqrt{T}$ (T in kelvin) and independent of pressure at constant temperature.

5 **Intensity** is the rate of energy flow per unit area perpendicular to the direction of propagation; $I = \frac{P}{A}$ and $I \propto (\text{amplitude})^2$.

6 Displacement nodes are pressure antinodes and displacement antinodes are pressure nodes; the pressure variation is $\frac{\pi}{2}$ out of phase with the displacement.

22.2 Sources of Sound

7 **Standing (stationary) waves** in air columns and strings are formed by the superposition of a progressive wave and its reflection, giving fixed nodes and antinodes.

8 **Resonance** occurs when the driving frequency equals a natural frequency of the system, producing a large-amplitude oscillation.

9 In an **open pipe** there is a displacement antinode at each end, so all harmonics are present: $f_n = \frac{nv}{2L}$.

10 In a **closed pipe** there is a displacement node at the closed end and an antinode at the open end, so only odd harmonics are present: $f_n = \frac{(2n - 1)v}{4L}$.

11 In a stretched string fixed at both ends there is a node at each end; the wave speed on the string is $v = \sqrt{\frac{T}{\mu}}$ and $f_n = \frac{nv}{2L}$.

12 The antinode at an open end lies slightly beyond the pipe; this **end correction** makes the effective length of the air column larger than the physical length.

22.3 Intensity Level of Sound

13 **Intensity level** (β) is a logarithmic measure of the intensity of a sound relative to the threshold of hearing: $\beta = 10 \log_{10}\frac{I}{I_0}$, measured in decibels (dB).

14 The **threshold of hearing** $I_0 = 1.0\times10^{-12}$ W m^{-2} is the minimum intensity audible at about 1 kHz.

15 A difference of 10 dB corresponds to a ratio of intensities of 10; a difference of about 3 dB corresponds to a ratio of 2.

16 Intensity levels in dB cannot be added directly; the intensities of incoherent sources add, and the total is then converted to dB.

17 For a point source the intensity obeys the inverse square law, $I \propto \frac{1}{r^2}$, because the same power spreads over a sphere of area $4\pi r^2$.

22.4 Beat

18 **Beats** are the periodic variations in intensity (loudness) caused by the superposition of two sound waves of slightly different frequencies.

19 Beats are produced because the two waves alternate between being in phase (constructive interference, loud) and in antiphase (destructive interference, soft).

20 The **beat frequency** is the number of loud (or soft) sounds heard per second and equals the magnitude of the difference between the two frequencies.

21 The frequency of the resultant sound is the mean of the two frequencies, and its amplitude varies at the beat frequency.

22 Beats can be used to compare an unknown frequency with a standard; loading or filing one source changes the beat frequency in a predictable way.

22.5 Doppler Effect

23 The **Doppler effect** is the change in the observed frequency of a wave due to relative motion between the source and the observer.

24 When a source moves towards a stationary observer, the wavelength is shortened and the observed frequency is higher; when it moves away, the wavelength is lengthened and the frequency is lower.

25 When an observer moves towards a stationary source, more wavefronts are intercepted per second, so the observed frequency is higher; moving away gives a lower frequency.

26 Speeds are taken relative to the air (medium); the speed of sound in air is unchanged by the motion of source or observer.

27 The Doppler shift of waves reflected from a moving object is used in radar speed guns and in medical blood-flow measurement.

CHAPTER 23 — GEOMETRICAL OPTICS

23.1 Spherical Mirrors

1 **Sign convention used throughout (real-is-positive):** distances of real objects and real images are positive; virtual images (behind the mirror) have negative v; f and R are positive for a concave mirror and negative for a convex mirror.

2 The **centre of curvature** C is the centre of the sphere of which the mirror is a part; the **radius of curvature** r is the radius of that sphere.

3 The **pole** P is the centre of the reflecting surface of the mirror; the **aperture** is the size (area) of the reflecting surface.

4 The **principal axis** is the line through the pole and the centre of curvature; it is normal to the mirror at its pole.

5 The **principal focus** of a concave mirror is the point on the principal axis through which rays parallel and close to the axis converge after reflection.

6 The **focal length** is the distance from the pole to the principal focus; $f = \frac{R}{2}$ for a spherical mirror of small aperture, where R is the radius of curvature.

7 **Linear magnification** is the ratio of the image height to the object height, $m = \frac{h_i}{h_o} = -\frac{v}{u}$; m is negative for an inverted (real) image and positive for an upright (virtual) image.

8 A concave mirror forms a real, inverted image for $u > f$ and a virtual, upright, magnified image for $u < f$; a convex mirror always forms a virtual, upright, diminished image.

23.2 Refraction at Spherical Surfaces

9 **Sign convention (real-is-positive):** u is positive for a real object in front of the surface; v is positive for a real image formed on the side where light emerges, negative for a virtual image on the object side; R is positive when the centre of curvature lies on the outgoing (emerging) side, i.e. a surface convex towards the incident light.

10 Refraction at a spherical surface obeys Snell's law $n_1 \sin i = n_2 \sin r$ at each point; for paraxial rays this leads to the surface equation below.

11 **Apparent depth**: an object in a denser medium appears closer to the surface when viewed normally, because rays bend away from the normal on leaving it.

12 A plane surface has $R \to \infty$, so the term $\frac{n_2 - n_1}{R}$ vanishes; then $\frac{n_1}{u} + \frac{n_2}{v} = 0$ and $\frac{v}{u} = -\frac{n_2}{n_1}$.

23.3 Thin Lenses

13 **Sign convention (real-is-positive):** u positive for a real object; v positive for a real image (on the opposite side of the lens), negative for a virtual image; f positive for a converging (convex) lens and negative for a diverging (concave) lens.

14 The **principal focus** of a converging lens is the point on the axis to which rays parallel and close to the axis converge after refraction through the lens.

15 The **power** of a lens is the reciprocal of its focal length in metres, $P = \frac{1}{f}$, with unit dioptre (D).

16 **Linear magnification** $m = \frac{h_i}{h_o} = -\frac{v}{u}$; a real image from a single lens is inverted (m negative); a virtual image is upright (m positive).

17 A convex lens forms a real, inverted image when u > f and a virtual, upright, magnified image when u < f (magnifying glass); a concave lens always forms a virtual, upright, diminished image.

CHAPTER 24 — WAVE OPTICS

24.1 Huygens's Principle

1 **Wavefront** is a line or surface joining all points of a wave that are in phase (same displacement and same direction of motion).

2 **Huygens's principle**: every point on a wavefront acts as a source of secondary spherical wavelets that spread out with the speed of the wave; the new wavefront after a short time t is the envelope (common tangent) of these wavelets.

3 **Diffraction** is the spreading of a wave into the geometrical shadow when it passes the edge of an obstacle or through a gap; the points at the edge act as sources of secondary wavelets that spread into the shadow region.

4 **Reflection and refraction**: construction with secondary wavelets gives the law of reflection (angle of incidence = angle of reflection) and Snell's law; on refraction the frequency is unchanged while the speed and wavelength change.

24.2 Interference

5 **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.

6 **Interference** is the superposition of two or more coherent waves, producing a pattern of reinforcement (maxima) and cancellation (minima).

7 **Coherent sources** are sources that emit waves of the same frequency (wavelength) with a constant phase difference.

8 **Optical path** between two points in a medium is the distance in vacuum that contains the same number of wavelengths as the medium between the two points, for light of the same frequency; optical path = nd, where n is the refractive index and d the actual distance.

9 **Constructive interference** occurs when the waves meet in phase (path difference = nλ, n = 0, 1, 2, ...); **destructive interference** occurs when they meet in antiphase (path difference = (n + ½)λ).

10 Two separate light bulbs do not give a stable interference pattern because the phase relationship of their light changes randomly and rapidly, so they are not coherent.

11 On reflection from a medium of higher refractive index (denser medium) the wave undergoes a phase change of π (equivalent to a path change of $\frac{\lambda}{2}$); there is no phase change on reflection at a less dense medium.

24.3 Two-slit Interference Pattern

12 **Young's double-slit experiment**: light from a single narrow slit (or a laser) illuminates two narrow slits S_1 and S_2 which act as coherent sources; the overlapping diffracted waves produce alternate bright and dark fringes on a screen.

13 The bright fringe is formed where the path difference $S_2P - S_1P = n\lambda$; the dark fringe where $S_2P - S_1P = (n + ½)\lambda$.

14 Fringes are equally spaced and of nearly equal brightness near the centre; the central fringe ($n = 0$) is bright because the path difference is zero.

15 With white light, the central fringe is white and the other fringes are coloured (violet nearest the centre, red furthest) since fringe spacing $\propto \lambda$.

24.4 Interference in A Thin Film

16 **Thin-film interference** is the interference between the light reflected from the top surface and the light reflected from the bottom surface of the film; the two beams are coherent because they originate from the same incident wave.

17 A phase change of π occurs on reflection when the light is reflected from a medium of higher refractive index; there is no phase change on reflection from a medium of lower refractive index.

18 **Colours of a thin film in white light**: the film thickness varies, so different wavelengths satisfy the condition for constructive interference at different places, giving coloured bands.

19 **Air wedge**: the air layer between two glass plates in contact at one edge; the fringes are straight, parallel and equally spaced, and the fringe at the line of contact is dark because of the π phase change at one reflection only.

20 **Non-reflective coating**: a thin film of refractive index between those of air and glass; the reflected beams from the two surfaces interfere destructively so that reflection is reduced.

24.5 Diffraction by A Single Slit

21 **Diffraction** is the spreading of waves when they pass through a gap or round an obstacle; it is most marked when the width of the gap is comparable with the wavelength.

22 A single-slit diffraction pattern has a wide, bright central maximum with weaker secondary maxima on both sides; the intensity of the secondary maxima falls rapidly.

23 The pattern arises from interference between the secondary wavelets from different points across the width of the slit.

24 The central maximum is twice as wide as each secondary maximum; the minima occur where the path difference between the two edges of the slit is a whole number of wavelengths.

24.6 Diffraction Gratings

25 **Diffraction grating** is an optical device with a large number of equally spaced parallel slits (lines) of very narrow width; the diffracted beams interfere to produce sharp, bright maxima at definite angles.

26 **Grating spacing d** is the distance between the centres of adjacent slits; $d = \frac{1}{N}$, where N is the number of lines per unit length.

27 The maxima occur where the path difference between light from adjacent slits is a whole number of wavelengths.

28 With white light, the zero-order maximum is white and each higher order is a spectrum with violet nearest and red furthest from the centre; the higher orders may overlap.

29 Increasing the number of lines illuminated makes the maxima sharper and brighter; the angular positions of the maxima depend on d and λ only.

24.7 Polarisation

30 **Plane-polarised (linearly polarised) light** is light in which the electric field vibrations are in one plane only, perpendicular to the direction of propagation.

31 **Unpolarised light** is light whose electric field vibrations occur in all planes perpendicular to the direction of propagation; it can be resolved into two mutually perpendicular components of equal intensity, both perpendicular to the direction of propagation.

32 **Plane of polarisation** is the plane containing the electric field vibrations of the wave and the direction of propagation.

33 Only transverse waves can be polarised; the polarisation of light shows that light is a transverse wave, whereas sound (a longitudinal wave) cannot be polarised.

34 **Polariser** produces plane-polarised light from unpolarised light; **analyser** is a second polariser used to detect the plane of polarisation.

35 **Malus's law**: the intensity of plane-polarised light transmitted through an analyser is proportional to the square of the cosine of the angle between the plane of polarisation of the light and the transmission axis of the analyser.

36 **Brewster's law**: when light is incident at the polarising angle θ_B, the reflected light is completely plane-polarised (vibrations perpendicular to the plane of incidence) and the reflected and refracted rays are perpendicular to each other.

37 Unpolarised light passing through an ideal polariser is reduced to half of the incident intensity.

24.8 Optical Waveguides

38 **Optical fibre** is a thin strand of glass or plastic consisting of a core surrounded by a cladding of lower refractive index; light is guided along the core by total internal reflection.

39 **Total internal reflection** occurs when light travels from a denser to a less dense medium and the angle of incidence exceeds the critical angle.

40 **Critical angle θ_c** is the angle of incidence in the denser medium for which the angle of refraction in the less dense medium is 90°.

41 Conditions for guiding: $n_{core} > n_{cladding}$ and the angle of incidence at the core–cladding boundary greater than θ_c.

42 **Attenuation** is the loss of power as the signal travels along the fibre, caused by absorption and scattering in the glass.

43 **Modal dispersion** is the spreading (broadening) of a pulse because rays following different paths take different times; it is reduced in a very narrow core (single-mode) fibre or a graded-index fibre.

CHAPTER 25 — QUANTUM PHYSICS

25.1 Photon

1 **Photon** is a quantum (packet) of electromagnetic radiation of energy E = hf; it is a neutral particle with zero rest mass that travels at the speed of light.

2 **Photoelectric effect** is the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency falls on it.

3 **Compton effect** is the scattering of X-ray photons by electrons in atoms; the scattered photon has a longer wavelength (lower energy) than the incident photon, showing that a photon carries momentum.

4 **Work function φ** is the minimum energy required to remove an electron from the surface of a metal; **threshold frequency f_0** is the minimum frequency for emission, $f_0 = \frac{\varphi}{h}$.

5 **Stopping potential V_s** is the minimum reverse potential difference that reduces the photocurrent to zero; $eV_s = K_{max}$.

6 **Einstein's photoelectric equation**: $hf = \varphi + K_{max}$, an electron absorbs one photon whose energy is shared between the work function and the kinetic energy.

7 Observations that wave theory cannot explain: a threshold frequency exists; K_{max} is independent of intensity and depends on frequency; emission is instantaneous.

25.2 Wave-Particle Duality

8 **Wave-particle duality**: radiation and matter exhibit both wave properties (diffraction, interference) and particle properties (photoelectric effect, photon momentum) depending on the experiment.

9 **de Broglie hypothesis**: a moving particle of momentum p has an associated wavelength $\lambda = \frac{h}{p}$.

10 Wave nature of electrons is shown by electron diffraction through a thin crystal or graphite film (Davisson–Germer experiment): a ring (diffraction) pattern is formed.

11 Particle nature of light is shown by the photoelectric effect; wave nature of light by diffraction and interference.

12 The wave behaviour of macroscopic objects is not observed because the de Broglie wavelength is far too small compared with any obstacle or gap.

25.3 Atomic Structure

13 **Line spectrum** consists of discrete bright lines (emission) or dark lines (absorption) of definite wavelengths, characteristic of the element; it shows that electrons in atoms occupy discrete energy levels.

14 **Bohr's postulates**: electrons move in certain stable orbits without radiating energy (stationary states); the angular momentum of an orbiting electron is quantised, $mvr = \frac{nh}{2\pi}$; radiation is emitted or absorbed only when an electron makes a transition between two levels, $hf = E_i - E_f$.

15 **Energy level** is the energy of an electron in an allowed orbit in the atom.

16 **Ground state** is the state in which the electron in the atom has the lowest energy (n = 1); an **excited state** is any state with energy higher than the ground state.

17 **Excitation energy** is the energy needed to raise an electron from the ground state to an excited state.

18 **Ionisation energy** is the minimum energy needed to remove an electron from the ground state of an atom so that it is completely detached from the atom ($n = \infty$); for hydrogen it is 13.6 eV.

19 **Spectral series of hydrogen**: Lyman (transitions to n = 1, ultraviolet), Balmer (to n = 2, visible), Paschen (to n = 3, infrared).

20 **Laser action**: stimulated emission produces photons identical in phase, frequency and direction to the stimulating photon; population inversion is needed for light amplification.

25.4 X-rays

21 **X-rays** are electromagnetic waves of very short wavelength (about 10^{-11} to 10^{-9} m), produced when fast electrons are suddenly stopped by a metal target.

22 **Continuous spectrum** arises from the deceleration of electrons in the target (bremsstrahlung); it has a sharp cut-off at the minimum wavelength λ_{min}, corresponding to the whole kinetic energy of an electron being converted to a single photon.

23 **Characteristic spectrum** consists of sharp lines produced when an incident electron ejects an inner-shell electron of the target atom and an outer electron falls to fill the vacancy; the lines are characteristic of the target material.

24 **Bragg's law**: constructive interference occurs for X-rays reflected from successive crystal planes when the path difference $2d \sin \theta$ is a whole number of wavelengths.

25 **Glancing angle** θ is the angle between the incident X-ray beam and the atomic (crystal) plane.

26 Intensity of the X-ray beam depends on the tube current; the penetrating power (hardness) depends on the accelerating voltage.

25.5 Nanoscience

27 **Nanoscience** is the study of materials and structures with at least one dimension of the order of 1 nm to 100 nm.

28 The surface-area-to-volume ratio of nanoparticles is very large, so a large proportion of atoms lie at the surface, which gives a high chemical reactivity and catalytic activity.

29 **Scanning tunnelling microscope (STM)** uses the quantum-mechanical tunnelling current between a sharp conducting tip and the surface; the current depends very sensitively (exponentially) on the tip–surface separation, so atoms can be imaged.

30 Examples: carbon nanotubes (high strength, electrical conduction), graphene, nanoparticles in sunscreens and catalysts.

CHAPTER 26 — NUCLEAR PHYSICS

26.1 Nucleus

1 The **proton (atomic) number** Z is the number of protons in the nucleus; the **nucleon (mass) number** A is the total number of protons and neutrons; the number of neutrons N = A − Z.

2 **Isotopes** are nuclides of the same element (same Z) with different numbers of neutrons (different A).

3 A **nuclide** is a particular species of nucleus with a specified proton number Z and nucleon number A.

4 The **unified atomic mass unit** (u) is one twelfth of the mass of a neutral carbon-12 atom; 1 u = 1.66×10^{-27} kg ≡ 931.5 MeV.

5 The **mass defect** is the difference between the total mass of the separate nucleons and the mass of the nucleus.

6 The **binding energy** is the energy required to separate a nucleus into its individual nucleons; $B = \Delta mc^2$.

7 The **binding energy per nucleon** is the binding energy of a nucleus divided by its number of nucleons. It is a measure of nuclear stability: a nucleus with a greater binding energy per nucleon is more stable; the curve rises steeply for light nuclei, peaks at about 8.8 MeV near ^{56}Fe (A ≈ 56) and falls slowly for heavy nuclei.

8 The nuclear radius $R = R_0A^{1/3}$, so the nuclear density is the same for all nuclei (about 2×10^{17} kg m^{-3}).

26.2 Radioactivity

9 **Radioactive decay** is the spontaneous and random disintegration of an unstable nucleus with the emission of α, β or γ radiation.

10 Radioactive decay is **random** and **spontaneous**. Random: it is impossible to predict when a particular nucleus will decay, and every nucleus has the same probability of decay per unit time. Spontaneous: the decay is not affected by external physical conditions such as temperature and pressure.

11 The **activity** A is the rate of decay of the nuclei, $A = -\frac{dN}{dt} = \lambda N$, measured in becquerel (Bq; 1 Bq = 1 decay per second).

12 The **decay constant** λ is the probability of decay per unit time of a nucleus.

13 The **half-life** $t_{½}$ is the time taken for the number of undecayed nuclei (or the activity) to fall to half of its initial value.

14 α emission reduces A by 4 and Z by 2; β^- emission raises Z by 1 with A unchanged; γ emission changes neither A nor Z.

26.3 Nuclear Reaction

15 In a nuclear reaction, mass number, charge (atomic number), energy and momentum are conserved; rest mass is not conserved.

16 The **Q-value** of a reaction is the energy released (Q > 0) or absorbed (Q < 0), Q = (total mass of reactants − total mass of products)c^2.

17 **Nuclear fission** is the splitting of a heavy nucleus (e.g. ^{235}U, after absorbing a neutron) into two lighter nuclei with the release of energy and neutrons; each fission of ^{235}U releases about 200 MeV.

18 **Nuclear fusion** is the joining of two light nuclei to form a heavier nucleus with the release of energy; it requires a very high temperature to overcome electrostatic repulsion.

19 Energy is released in fission and fusion because the products have a greater binding energy per nucleon, so their total mass is less than that of the reactants.

20 A **chain reaction** is sustained when, on average, at least one of the secondary neutrons from each fission causes another fission.

21 A **thermal neutron** is a slow neutron whose kinetic energy is about the same as the thermal energy of particles at room temperature (about 0.025 eV); thermal neutrons are the most likely to cause fission of ^{235}U.

22 A **thermonuclear reaction** is a fusion reaction that occurs when the temperature of the light nuclei is raised to about 10^8 K, so that they collide with enough kinetic energy to overcome the electrostatic repulsion and fuse.